STPM 2026 PHYSICS 960/1

Questions and Answers

All calculations use $g = 9.81\ m\ s^{-2}$ and the values of constants printed in the question paper.

Section A [20 marks]

1. Two forces, 25.0 N and F, act on one point object. If the resultant force is 84.0 N at an angle of 30.0° from 25.0 N, what is the magnitude of F?

A 14.4 N B 63.6 N C 74.7 N D 80.2 N

Answer: B

Take the 25.0 N force along the x-axis. F = resultant − 25.0 N (vector subtraction):

$$F_x = 84.0 \cos 30.0° - 25.0 = 47.7\ \text{N}$$

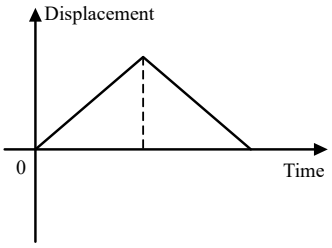
$$F_y = 84.0 \sin 30.0° = 42.0\ \text{N}$$

$$F = \sqrt{(47.7^2 + 42.0^2)} = 63.6\ \text{N}$$

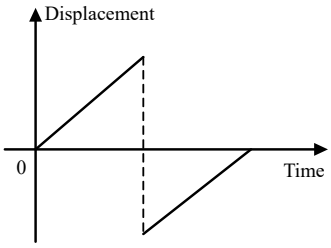
Explanation: The resultant is the vector sum of the two forces, so the unknown force is found by subtracting vectors, component by component. Subtracting the magnitudes (84.0 − 25.0) only works when the forces are parallel. Option D (80.2 N) comes from assuming F is perpendicular to the 25.0 N force: $\sqrt{(84.0^2 - 25.0^2)}$.

2. An object is thrown vertically upwards and reaches the maximum height before it falls to the ground. Which displacement-time graph describes the motion of the object?

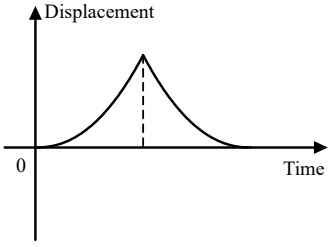
A



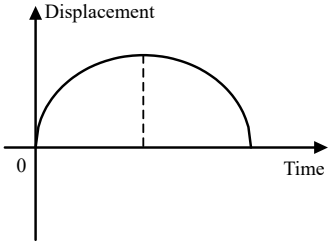
B



C



D



Answer: D

Displacement increases to a maximum and then decreases. The velocity (the gradient of the graph) decreases uniformly from positive, through zero at the maximum height, to negative, so the graph is a smooth curve (parabola) that is concave downward.

$$s = ut - \frac{1}{2}gt^2$$

Explanation: The gradient of a displacement–time graph is the velocity. Under constant acceleration g the gradient must change smoothly, so the graph cannot have a sharp corner (A) or a sudden jump (B). Graph C has the gradient increasing, which would mean the object speeds up on the way up.

3. A car is moving in a straight line with a stationary small ball hung on the rear view mirror by a string. Which motion of the car causes the small ball to swing towards the front of the car?

A Accelerating forward
B Decelerating in forward motion
C Moving forward with constant velocity
D Moving backward with constant velocity

Answer: B

When the car decelerates, the ball tends to continue moving forward at the original speed because of its inertia (Newton's first law), so it swings towards the front of the car.

Explanation: The string only pulls the ball when the car changes its velocity. At constant velocity (C and D) the string hangs vertically. When the car accelerates forward (A) the ball lags behind and swings towards the back; when the car slows down (B) the ball keeps going and swings towards the front.

4. A man, P, of mass 50 kg climbs up a staircase from the first floor to the second floor which is 5.0 m above the first floor. He takes 2.5 minutes to reach the second floor. Another man, Q, takes the same time to climb up the same staircase eventhough his mass doubles the mass of P. Which statement is **not** true about the power of Q?

A The power of Q is 32.7 W.
B Q has greater power than P.
C The power of Q is directly proportional to the mass of Q.
D Q has to climb faster to the second floor to obtain the same power of P.

Answer: D

$$\text{Power} = \frac{mgh}{t}$$

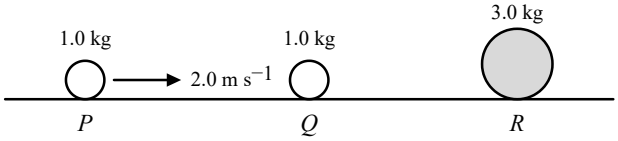
$$\text{Power of } P = \frac{(50)(9.81)(5.0)}{150} = 16.4 \text{ W}$$

$$\text{Power of } Q = \frac{(100)(9.81)(5.0)}{150} = 32.7 \text{ W}$$

A, B and C are true. Since Q does twice the work of P in the same time, Q already has twice the power of P, so D is not true.

Explanation: Power is the rate of doing work against gravity, $P = \frac{mgh}{t}$. With the same height and the same time, power is directly proportional to mass. To have the same power as P, Q would have to climb *more slowly* (take twice as long), not faster. Remember to convert 2.5 minutes to 150 s.

5. Two identical steel balls of mass 1.0 kg, P and Q, and another steel ball of mass 3.0 kg, R, are placed on a smooth surface of a table as shown in the diagram below.



P moves with a velocity of 2.0 m s^{-1} and collides elastically with Q which is stationary. Q then collides perfectly inelastic with R and moves with a common velocity. What is the velocity of R after the collision?

A 0.25 m s^{-1} B 0.50 m s^{-1} C 0.67 m s^{-1} D 1.0 m s^{-1}

Answer: B

P and Q have equal masses and collide elastically, so they exchange velocities: Q moves off with 2.0 m s^{-1}.

Q and R (perfectly inelastic collision):

$$m_Q u_Q = (m_Q + m_R)v$$
$$(1.0)(2.0) = (1.0 + 3.0)v$$
$$v = 0.50 \text{ m s}^{-1}$$

Explanation: In a head-on elastic collision between equal masses, one of which is at rest, the moving ball stops and the other moves off with its velocity. In a perfectly inelastic collision the bodies stick together and move with a common velocity: momentum is conserved but kinetic energy is not.

6. A girl sits on a merry-go-round which is in motion. Where should the girl sit so that she experiences the largest possible centripetal force?

A Close to the edge
B Close to the centre
C Any distance from the centre
D Midpoint between the centre and the edge

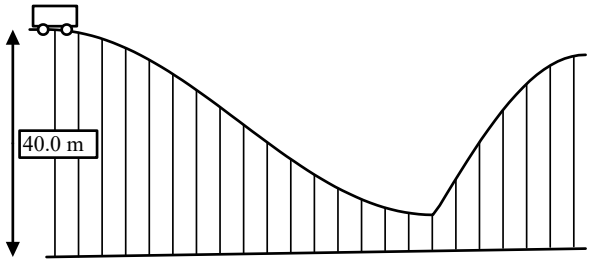
Answer: A

$$F = m\omega^2 r$$

All points on the merry-go-round have the same angular velocity ω, so $F \propto r$. The force is largest at the largest radius, i.e. close to the edge.

Explanation: Every point of a rotating rigid platform completes one revolution in the same time, so ω is the same everywhere, but the linear speed $v = \omega r$ is greatest at the edge. Using $F = \frac{mv^2}{r}$ alone is misleading because v also changes with r; $F = m\omega^2 r$ shows the dependence clearly.

7. A roller coaster cart on a track moves from the maximum height of 40.0 m as shown in the diagram below.



If the friction on the track is neglected, what is the centripetal acceleration experienced by the cart when passing the lowest point of the curve which has a radius of 15.0 m?

A $19.6\ \text{m s}^{-2}$ B $26.2\ \text{m s}^{-2}$ C $32.7\ \text{m s}^{-2}$ D $52.3\ \text{m s}^{-2}$

Answer: D

Conservation of energy (lowest point at ground level):

$$\frac{1}{2}mv^2 = mgh \Longrightarrow v^2 = 2gh = 2(9.81)(40.0) = 784.8\ \text{m}^2\ \text{s}^{-2}$$

$$a = \frac{v^2}{r} = \frac{784.8}{15.0} = 52.3\ \text{m s}^{-2}$$

Explanation: With no friction, the loss of gravitational potential energy from the top becomes kinetic energy at the bottom; the mass cancels. The centripetal acceleration $\frac{v^2}{r}$ points towards the centre of the curve, i.e. upwards at the lowest point. Option B ($26.2\ \text{m s}^{-2}$) comes from forgetting the factor 2 in $v^2 = 2gh$.

8. Two planets, P and Q, with an equal density have the radii of r and $2r$ respectively. If the gravitational field strength on the surface of P is g, what is the gravitational field strength on the surface of Q?

A $\frac{1}{8}g$ B $\frac{1}{2}g$ C g D $2g$

Answer: D

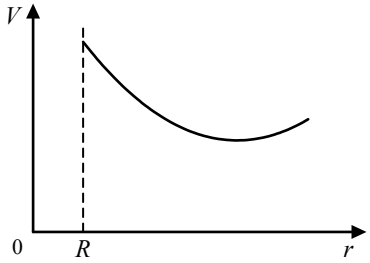
$$g = \frac{GM}{r^2} = \frac{G(\frac{4}{3}\pi r^3 \rho)}{r^2} = \frac{4}{3}\pi G \rho r$$

For planets of equal density, $g \propto r$. The radius of Q is $2r$, so the field strength on Q is $2g$.

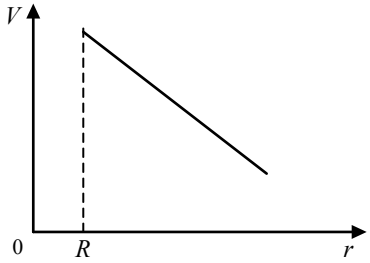
Explanation: The planets have the same *density*, not the same mass. Doubling the radius makes the mass $2^3 = 8$ times larger, while the distance to the centre is only doubled, so $g = \frac{8GM}{(2r)^2} = 2 \times \frac{GM}{r^2}$. Option A ($\frac{1}{8}g$) and B wrongly keep the mass the same or mix up the powers.

9. The gravitational potential, V, at a point varies with distance, r, from the centre of the Earth, where r is bigger than the radius of the Earth, R. Which graph represents the variation of V with r?

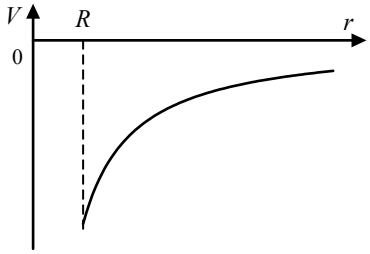
A



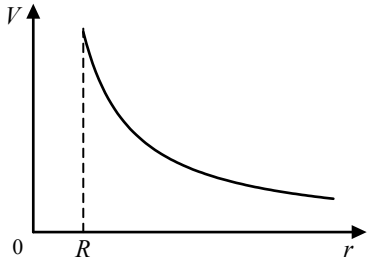
B



C



D



Answer: C

$$V = -\frac{GM}{r}$$

V is negative for all $r > R$ and increases (becomes less negative) towards zero as r increases.

Explanation: Gravitational potential is defined as zero at infinity. Because gravity is attractive, work must be done to move a mass out to infinity, so the potential everywhere else is negative. Its magnitude is proportional to $\frac{1}{r}$, so the graph rises towards zero. Graph D has the right shape but positive values.

10. Which statement is true about the centre of gravity?

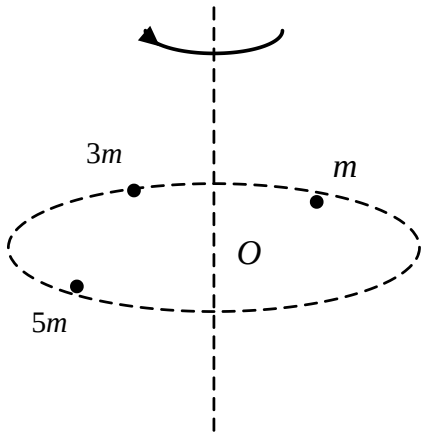
A The centre of gravity of a uniform disc lies anywhere on its circumference.
B The centre of gravity of an object is equal to its centre of mass in uniform gravitational field.
C The centre of gravity of a uniform cylinder lies along the midpoint of its cross-sectional area.
D The centre of gravity of an object at North pole is different from the centre of gravity of the same object at the equator.

Answer: B

In a uniform gravitational field, g is the same at every point of the object, so the centre of gravity coincides with the centre of mass.

Explanation: The centre of mass is where the mass of a body can be considered to be concentrated; the centre of gravity is where its whole weight can be considered to act. They differ only if g changes across the body. For everyday objects g is effectively uniform, so they are at the same point (and this is true at the pole as well as at the equator, so D is wrong). The centre of gravity of a uniform disc is at its centre, not on the circumference.

11. Three point masses, m, $3m$ and $5m$, are located on the same plane at the same distance, r, from O are shown in the diagram below.



What is the moment of inertia of the system about the axis of rotation through O?

A $\frac{9}{2}mr^2$ B $7mr^2$ C $9mr^2$ D $35mr^2$

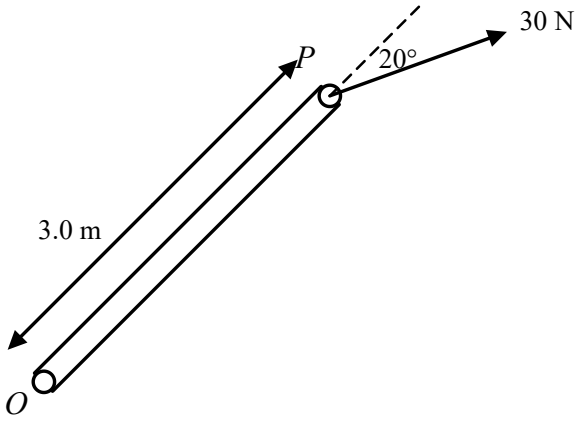
Answer: C

$$I = \Sigma mr^2 = mr^2 + 3mr^2 + 5mr^2 = 9mr^2$$

All three masses are at the same distance r from the axis through O.

Explanation: The moment of inertia of point masses is the sum of each mass times the *square* of its perpendicular distance from the axis. The positions of the masses around the circle do not matter, only their distance from the axis.

12. A rod of length 3.0 m is pivoted at O. A force of 30 N is acted on P with an angle of 20° to the axis of the rod as shown in the diagram below.



What is the torque due to the force about O?

A 31 N m B 85 N m C 90 N m D 92 N m

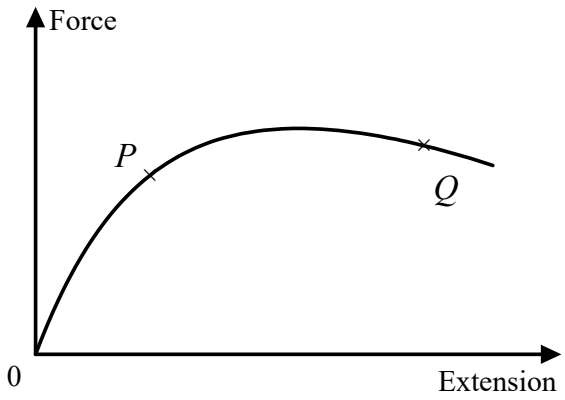
Answer: A

Only the component of the force perpendicular to the rod produces a torque about O.

$$\tau = Fr \sin\theta = (30)(3.0) \sin 20° = 31 \text{ N m}$$

Explanation: Torque = force × perpendicular distance from the pivot to the line of action of the force. The component of the force along the rod (30 cos 20°) passes through O and has no turning effect. Option C (90 N m) assumes the force is perpendicular to the rod; option B (85 N m) uses cos 20° instead of sin 20°.

13. The variation of force with an extension of a material is shown in the graph below.



Which statement is **not** true about the material?

A The material breaks when it passes Q.
B The area under the curve OPQ represents the strain energy in the material.
C The material experiences permanent extension when it is stretched more than P.
D The strain of the material is directly proportional with the stress before it reaches P.

Answer: D

The graph from *O* to *P* is a **curve**, not a straight line, so the extension (strain) is not directly proportional to the force (stress) before *P*. Statement D is not true.

P is the elastic limit (beyond it the extension is permanent, C) and *Q* is the breaking point (A). The area under a force–extension graph is the work done in stretching the material, i.e. the strain energy (B).

Explanation: Hooke's law (force ∝ extension, stress ∝ strain) holds only where the graph is a straight line through the origin. Here the line bends from the start, so there is no proportional region before *P*. Note: if *OP* were a straight line, D would be true, and B would then be the statement to question, because only the elastic part of the work done is stored as recoverable energy; read the shape of the graph carefully.

14. Which statement is true about real gas?

A Real gas behaves like ideal gas by obeying the gas equation $PV = nRT$.
B Real gas behaves like ideal gas at high pressure and low temperature.
C The internal energy of real gas consists of kinetic energy only.
D The intermolecular force in real gas is negligible.

Answer: A

A real gas behaves like an ideal gas when it obeys $pV = nRT$, i.e. at low pressure and high temperature (B is wrong). Real gas molecules have intermolecular forces, so the internal energy includes potential energy (C and D are wrong).

Explanation: At low pressure the molecules are far apart, so their own volume and the forces between them are negligible; at high temperature their kinetic energy is much larger than the potential energy of attraction. Under these conditions a real gas obeys the ideal gas equation.

15. A monatomic ideal gas of 1.5 moles is heated at constant pressure so that the temperature increases by 20 °C. What is the change in the internal energy of the gas?

A 3.7×10^2 J B 6.2×10^2 J C 5.5×10^3 J D 9.1×10^3 J

Answer: A

The change in internal energy depends only on the change in temperature:

$$\Delta U = nC_V\Delta T = n(\tfrac{3}{2}R)\Delta T$$

$$\Delta U = (1.5)(\tfrac{3}{2})(8.31)(20) = 3.7 \times 10^2 \text{ J}$$

Explanation: For an ideal gas the internal energy depends only on temperature, so $\Delta U = nC_V\Delta T$ for *any* process, even at constant pressure. For a monatomic gas $C_V = \frac{3}{2}R$. A change of 20 °C is the same as a change of 20 K. Option B (6.2×10^2 J) is the heat supplied, $nC_p\Delta T$, which is larger because the gas also does work as it expands.

16. The heat capacity for 400 g of copper is 156 J K^{-1}. What is the molar heat capacity for copper? [Molar mass for copper is 63.5 g mol^{-1}.]

A 2.46 J mol^{-1} K^{-1} B 9.91 J mol^{-1} K^{-1} C 24.8 J mol^{-1} K^{-1} D 62.4 J mol^{-1} K^{-1}

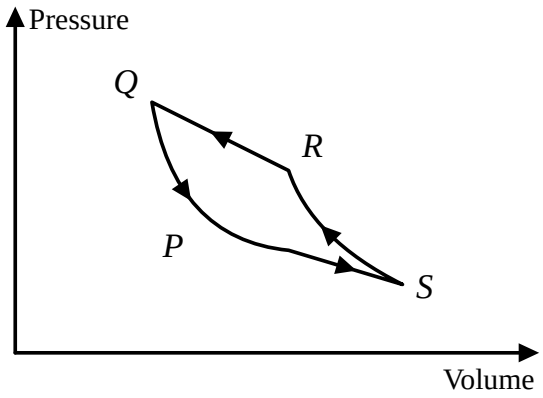
Answer: C

$$n = \frac{m}{M} = \frac{400}{63.5} = 6.30 \text{ mol}$$

$$C_m = \frac{C}{n} = \frac{156}{6.30} = 24.8 \text{ J mol}^{-1} \text{ K}^{-1}$$

Explanation: Heat capacity (J K^{-1}) belongs to the whole sample. Divide by the number of moles to get the molar heat capacity, or by the mass in kg to get the specific heat capacity (390 J kg^{-1} K^{-1}). Most solid metals have a molar heat capacity close to 25 J mol^{-1} K^{-1}, a useful check.

17. The variation of pressure with respect to volume of a gas is shown in the graph below.



Which statement represents the shaded area in the above graph?

A Work done on the gas during the changes of *SR* and *RQ*.
B Work done by the gas during the changes of *QP* and *PS*.
C Nett work done on the gas in the process of *SRQPS*.
D Nett work done by the gas in the process of *SRQPS*.

Answer: C

The cycle $S \rightarrow R \rightarrow Q \rightarrow P \rightarrow S$ is anticlockwise. The gas is compressed along $S \rightarrow R \rightarrow Q$ at higher pressure and expands along $P \rightarrow S$ at lower pressure, so the work done on the gas is greater than the work done by the gas. The enclosed area is the nett work done on the gas.

Explanation: The area under a process on a *p*–*V* graph is the work done; the area enclosed by a complete cycle is the *nett* work. A clockwise cycle gives nett work done *by* the gas (an engine); an anticlockwise cycle gives nett work done *on* the gas (a refrigerator or heat pump).

18. Three moles of an ideal gas in a cylinder is compressed at a constant temperature of 70.0 °C until the pressure becomes three times the original pressure. What is the work done by the process?

A −9390 J B −1920 J C 1920 J D 9390 J

Answer: A

$$W = nRT \ln \frac{V_2}{V_1} = nRT \ln \frac{p_1}{p_2}$$

$$W = (3)(8.31)(343) \ln \frac{1}{3} = -9390 \text{ J}$$

The negative sign shows that work is done on the gas (compression).

Explanation: At constant temperature pV is constant (Boyle's law), so tripling the pressure reduces the volume to one third. The work done *by* the gas is negative because the gas is compressed. The temperature must be in kelvin (70.0 + 273 = 343 K); option B (−1920 J) comes from using 70.0 instead of 343.

19. Which statement is true about natural convection and forced convection?

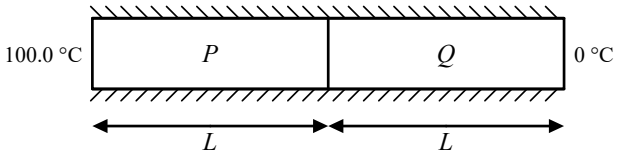
	Natural convection	*Forced convection*
A	The movement of fluid is due to density gradients within the fluid.	The fluid is forced to flow by an internal factor.
B	There is no external device or phenomenon which causes the fluid motion.	Typically the rate of heat transfer is relatively lower than natural convection for the same fluid.
C	There is external device or phenomenon which causes fluid motion.	The fluid is forced to flow by an external factor.
D	The movement of fluid is due to density gradients within the fluid.	Typically the rate of heat transfer is relatively higher than natural convection for the same fluid.

Answer: D

Natural convection is caused by density differences (buoyancy) within the fluid, while forced convection uses an external device (e.g. a fan or pump) and typically gives a higher rate of heat transfer.

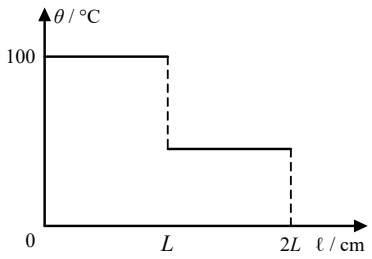
Explanation: In natural convection warmer fluid expands, becomes less dense and rises, while cooler fluid sinks. In forced convection a fan or pump moves the fluid faster, so warm fluid near the surface is replaced more quickly and the rate of heat transfer is higher. The fluid in forced convection is moved by an *external* factor, so A and B are wrong, and C describes natural convection wrongly.

20. Two well insulated metal rods, P and Q, with an equal cross-sectional area are joined end to end as shown in the diagram below.

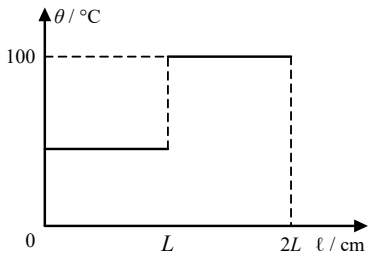


P has higher thermal conductivity than Q. Which graph shows the variation of temperature, θ, with length, ℓ, of P and Q?

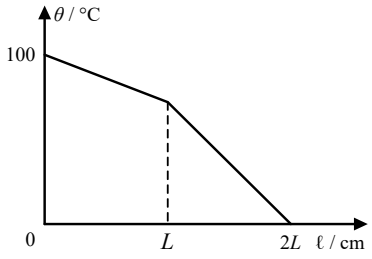
A



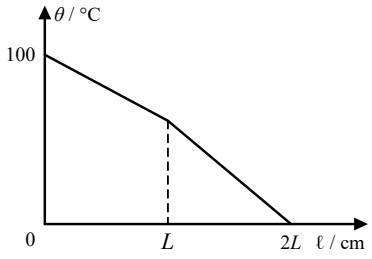
B



C



D



Answer: C

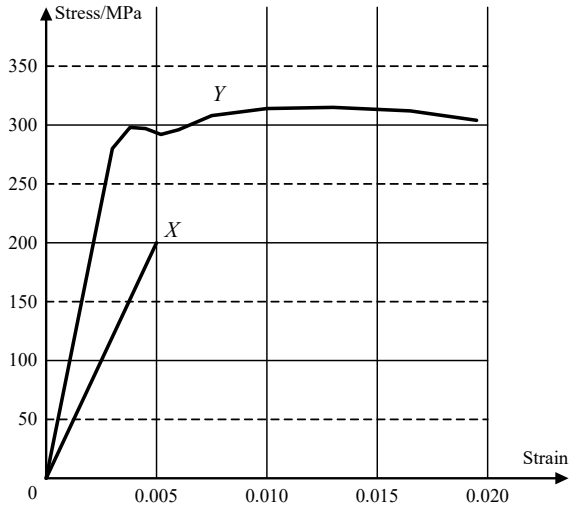
$$\frac{Q}{t} = kA\frac{\Delta\theta}{\Delta x}$$

The rate of heat flow and A are the same for both rods, so $\frac{\Delta\theta}{\Delta x} \propto \frac{1}{k}$. P has a higher k, so the temperature gradient in P is smaller and in Q is steeper; θ falls linearly from 100 °C to 0 °C with a change of gradient at the junction.

Explanation: The rods are insulated and joined in series, so in the steady state the same rate of heat flow passes through each rod. A good conductor needs only a small temperature gradient to carry this heat, while a poor conductor needs a steep gradient. Within each rod the temperature falls linearly, so the graph is two straight lines with the shallower one in P.

Section B [14 marks]

21. The stress-strain relationship of two materials, X and Y, are shown in the graph below.



(a) State the properties of *X* and *Y*. [2]

Answer

• ***X* is a brittle material**: it obeys Hooke's law right up to the breaking point with no plastic deformation (no yield point). 1

• ***Y* is a ductile material**: it has a yield point and undergoes large plastic deformation before it breaks. 1

2

Explanation: Look at what happens after the straight-line region. *X* ends suddenly at the end of its straight line (it breaks at a strain of 0.005 with no plastic region), so it is brittle. *Y* levels off at about 300 MPa and continues to a strain of 0.020: a large plastic deformation, so it is ductile.

(b) Determine the Young's modulus for *Y*. [2]

Answer

$E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\varepsilon}$ 1

$$E = \frac{270 \times 10^6}{0.0030}$$

$E = 9.0 \times 10^{10}$ Pa 1

[Accept 9.00×10^{10} Pa or 9.000×10^{10} Pa; prefix units may be used]

2

Explanation: The Young's modulus is the gradient of the straight-line (Hooke's law) part of the stress–strain graph. Choose a point on the straight line that can be read accurately, and convert MPa to Pa ($\times 10^6$).

(c) Estimate the strain energy per unit volume of *Y* until the yield point. [3]

Answer

Yield point: strain = 0.0040 – 0.0045 (any value in the range) 1

Energy per unit volume = area under the graph 1

$$W = \frac{1}{2}(0.0030)(270 \times 10^6) + \frac{1}{2}(270 + 300)(10^6)(0.0010)$$

$W = 6.9 \times 10^5$ J m^{-3} 1

OR (taking the yield point at strain 0.0045)

$$W = \frac{1}{2}(0.0030)(270 \times 10^6) + \frac{1}{2}(270 + 300)(10^6)(0.0015) = 8.3 \times 10^5 \text{ J m}^{-3}$$

[Accept 6.9×10^5, 6.90×10^5 or 6.900×10^5 J m^{-3} (or 8.3×10^5 J m^{-3} accordingly). No mark if only the strain energy equation is used without referring to the graph.]

3

Total for question **7**

Explanation: The area under a stress–strain graph is the work done (strain energy) per unit volume, because stress × strain = $\frac{F}{A} \times \frac{e}{l} = \frac{Fe}{Al}$. Split the area up to the yield point into a triangle (the linear part) and a trapezium. The answer depends on where you read the yield point, so a range is accepted.

22. A 60 W spherical frosted light bulb of radius 4.00 cm is made from glass of thickness 1.00 mm. Assume that the heat transfer through convection is negligible.

(a) Calculate the temperature of the outer surface of the bulb assuming the bulb is a black body radiating heat to the surrounding. [3]

Answer

$A = 4\pi r^2 = 4\pi(0.0400)^2 = 2.01 \times 10^{-2}\ \text{m}^2$ 1

$P = \sigma A T^4 \Longrightarrow T^4 = \frac{P}{\sigma A} = \frac{60}{(5.67 \times 10^{-8})(2.01 \times 10^{-2})}$ 1

$T = 479\ \text{K}$ 1

[Accept 480 K or 206 °C]

3

Explanation: In the steady state all 60 W supplied to the bulb leaves the outer surface as radiation. For a black body the Stefan–Boltzmann law gives $P = \sigma A T^4$, where A is the surface area of the sphere, $4\pi r^2$. Take the fourth root at the end.

(b) Calculate the temperature of the inner surface of the bulb if heat is transferred through the glass of the bulb by conduction only. [2]

Answer

$\frac{Q}{t} = kA\frac{(\theta_1 - \theta_2)}{x}; 60 = (0.300)(2.01 \times 10^{-2})\frac{(\theta_1 - 479)}{1.00 \times 10^{-3}}$ 1

$\theta_1 = 479 + 9.95 = 489\ \text{K}$ 1

[Accept 490 K or 216 °C; ECF from (a)]

2

Explanation: The same 60 W is conducted through the glass. Rearranging the conduction equation gives the temperature difference across the glass, $\frac{Px}{kA} \approx 10$ K. Heat flows outwards, so the inner surface is the hotter one. The glass is thin, so the outer surface area can be used.

(c) State whether the actual temperature of the outer surface of the bulb higher or lower than the value calculated in (a). Justify your answer.
[Thermal conductivity of glass is 0.300 W m^{-1} K^{-1}.] [2]

Answer

The actual temperature of the outer surface is **higher** than the calculated value.	1
• Because the bulb is not a black body (not a perfect emitter, $e < 1$).	1
	2
Total for question	**7**

Explanation: A real surface radiates $P = e\sigma AT^4$ with emissivity $e < 1$. To radiate the same 60 W with a smaller e, its temperature must be higher than the black-body value.

Section C [26 marks]

23. (a) State Newton's second and third laws of motion. [2]

Answer

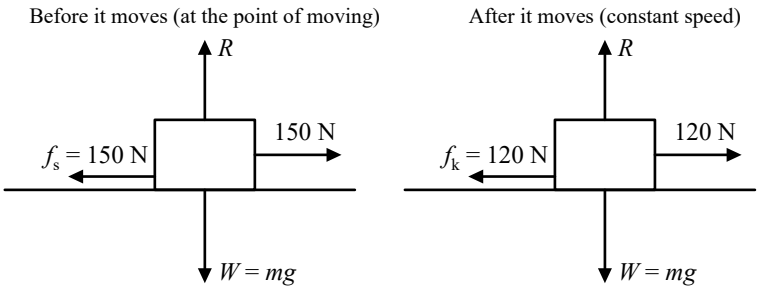
• Second law: the rate of change of momentum of a body is directly proportional to the resultant force acting on it and takes place in the direction of the force.	1
• Third law: if body A exerts a force on body B, then body B exerts a force of equal magnitude but in the opposite direction on body A.	1
	2

Explanation: The second law is about the *rate of change of momentum*; $F = ma$ is the special case for constant mass. The two forces in the third law act on *different* bodies, so they never cancel each other.

(b) A man pushes a box of mass 120 kg on a rough surface. The box starts to move when a force of 150 N is applied. The box continues moving at a constant speed with a force of 120 N.

(i) Sketch **two** free body diagrams of the box before and after it moves. [3]

Answer



• Weight, $W = mg$ (downwards) and normal reaction, R (upwards) shown on at least one diagram.	1
• Before it moves: applied force 150 N and maximum static friction f_s = 150 N in the opposite direction.	1
• After it moves: applied force 120 N and kinetic friction f_k = 120 N in the opposite direction.	1
	3

Explanation: A free body diagram shows every force acting on the box only. At the point of moving, static friction has reached its maximum value and just balances the 150 N push. At constant speed the resultant force is zero, so kinetic friction equals the 120 N push. Vertically, R balances the weight.

(ii) Calculate the coefficients of static and kinetic frictions. [5]

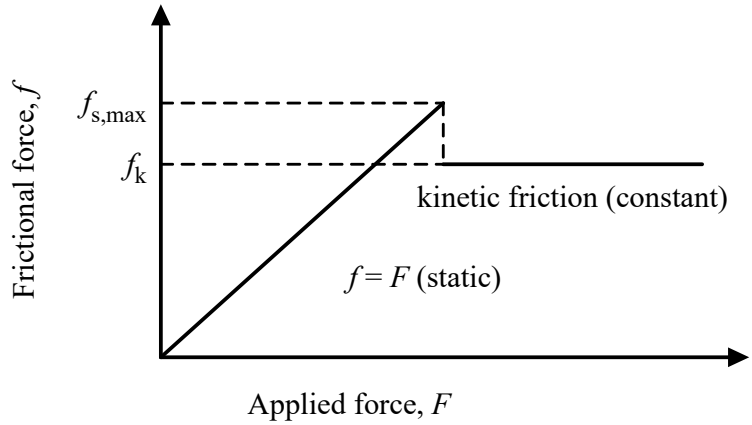
Answer

$\mu_s = \frac{f_s}{R}$	1
$W = R = mg$	1
$\mu_s = \frac{150}{120(9.81)} = 0.127$	1
$\mu_k = \frac{f_k}{R}$	1
$\mu_k = \frac{120}{120(9.81)} = 0.102$	1
[Accept μ_s = 0.13 or 0.1274 and μ_k = 0.10 or 0.1019]	
	5

Explanation: On a horizontal surface with a horizontal push, the normal reaction equals the weight. The coefficient of static friction is always larger than the coefficient of kinetic friction: it is harder to start an object moving than to keep it moving.

(c) A force is applied to a block that is initially at rest on a rough surface. The block moves and begins to accelerate as the applied force increases. Sketch a graph of frictional force against force that acts on the block. [3]

Answer



• Shape for static friction: straight line from the origin (frictional force = applied force) up to f_s.	1
• Shape for kinetic friction: drops from f_s to f_k and remains constant as the applied force increases.	1
• Labels for the frictional forces f_s and f_k.	1
[The graph must start from the origin]	
	3
Total for question	**13**

Explanation: While the block is at rest, static friction adjusts itself to equal the applied force, so the graph is a straight line through the origin with gradient 1. It can only increase up to its maximum value. Once the block slides, kinetic friction takes over: it is smaller and does not depend on the applied force, so the line is horizontal while the block accelerates.

24. (a) State **two** differences between thermal conduction and radiation. [2]

Answer

• Conduction requires a medium (direct contact); radiation does not require any medium and can occur in a vacuum.	1
• In conduction, heat is transferred through the vibration of atoms; in radiation, heat is transferred in the form of electromagnetic waves.	1
• Conduction occurs when there is a temperature difference between two points; radiation occurs when the temperature of an object is greater than 0 K.	1
• The rate of heat transfer by conduction is slow; the rate of heat transfer by radiation is fast.	1
[Any two – maximum 2 marks]	
	2

Explanation: Compare the two processes point by point: whether a medium is needed, how the energy is carried, when it happens and how fast. Radiation is the only process that can transfer heat through a vacuum, which is how energy reaches the Earth from the Sun.

(b)

(i) Explain greenhouse effect caused by the surface of the Earth. [3]

Answer

• The Earth's surface absorbs short-wavelength radiation (visible light) from the Sun and becomes warm.	1
• The warm surface of the Earth re-emits the energy as long-wavelength infrared radiation.	1
• Greenhouse gases (CO_2, water vapour, methane) absorb the infrared radiation and re-radiate it back to the Earth, trapping the heat and raising the temperature of the Earth.	1
	3

Explanation: The key idea is the change of wavelength. Short-wavelength sunlight passes through the atmosphere easily. The Earth is much cooler than the Sun, so it emits long-wavelength infrared radiation, and it is this infrared that greenhouse gases absorb and send back down.

(ii) State the impact of greenhouse effect on the Earth. [1]

Answer

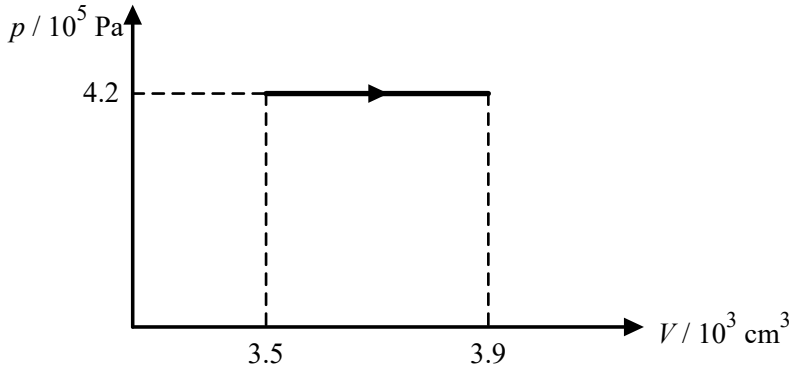
• Global warming: the average temperature of the Earth rises, causing melting of polar ice, rising sea level or climate change (any one).	1
	1

Explanation: An increase in greenhouse gases traps more infrared radiation, so the average temperature of the Earth rises.

(c) A constant mass of an ideal gas has a volume of 3.5×10^3 cm^3. The gas is then supplied with a heat of 565 J to expand under a constant pressure of 4.2×10^5 Pa. The final volume of the gas is 3.9×10^3 cm^3.

(i) Sketch a p-V graph of the process. [2]

Answer



• Correct shape: horizontal straight line (constant pressure) from V_i to V_f. 1

• Correct arrow in the direction of expansion (from V_i to V_f). 1

[Arrow in the opposite direction: no mark for the arrow. No arrow: 1 mark for the correct shape only. Values on the axes are not required.]

2

Explanation: A process at constant pressure (isobaric) is a horizontal line on a p–V graph. The gas expands, so the arrow points towards increasing volume.

(ii) Calculate the work done by the gas. [2]

Answer

$W = p\Delta V = (4.2 \times 10^5)(3.9 \times 10^{-3} - 3.5 \times 10^{-3})$ 1

$W = 168$ J 1

[Accept 170 J or 168.0 J]

2

Explanation: At constant pressure the work done by the gas is $p\Delta V$, the area under the horizontal line. Convert cm^3 to m^3 by multiplying by 10^{-6}: 3.9×10^3 cm^3 = 3.9×10^{-3} m^3.

(iii) Determine the change of internal energy of the gas. [2]

Answer

$Q = \Delta U + W \Longrightarrow \Delta U = Q - W = 565 - 168$ 1

$\Delta U = 397$ J 1

[Accept 400 J or 399.0 J; ECF from (ii)]

2

Explanation: By the first law of thermodynamics, part of the 565 J supplied is used by the gas to do 168 J of work as it expands; the rest increases the internal energy, so the temperature of the gas rises.

(iv) The gas is then compressed under isothermal process, what will happen to the final pressure of the gas? [1]

Answer

• The pressure increases: for an isothermal process pV = constant, so the pressure increases as the volume decreases (Boyle's law). 1

[Accept: higher]

1

Total for question **13**

Explanation: At constant temperature the pressure is inversely proportional to the volume. Compressing the gas makes the molecules hit the walls more often, so the pressure increases.